



Fat snorers and sleepy-heads: Were many distinguished characters of the past affected by the obstructive sleep apnea syndrome?

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Summary The obstructive sleep apnea syndrome (OSAS) is a clinical condition characterized by the coexistence of irregular breathing at night with excessive daytime sleepiness, and it represents a major social health issue because of its high prevalence and of the growing public awareness of it.

The XIX century description of “Fat Joe”, the famous character of Charles Dickens’s *Pickwick Papers*, is often retained the first presentation of a person affected by OSAS, since Joe was an obese individual who fell asleep during daytime while performing even extremely simple tasks.

However, apart from the fact that Joe’s Picwickian syndrome (“the obesity hypoventilation syndrome”) needs to be differentiated from true OSAS, already in the Hippocratic Corpus (V–IV century BC) many clues regarding apnea and sleepiness are present, and in 79 AD the Roman author Pliny the Younger clearly reported the death of a man in whom obesity, sleepiness and snoring were co-present in a unique clinical picture.

In this paper, elements suggesting OSAS are traced back and evidenced in the XIX and XX centuries in distinguished figures, including Emperor Napoleon Bonaparte, Queen Victoria and President Franklin D. Roosevelt. A posteriori reconstruction of the health status of characters of the past is difficult, and, to an extent, speculative, but the elements here provided appear relevant, since the possible presence of disabling OSAS in important personages of the past may have negatively influenced not only their health status, but also their social life and their work activity.

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Historical introduction

In 1836–1837 Charles Dickens wrote a series of stories about the Pickwick Club, stories that were

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later published as a novel entitled *The Pickwick Papers* [1]. “Fat Joe” is one of the most frequently cited characters of this book, mainly because of his physical constitution. In effect, Fat Joe is an obese fellow who often falls asleep not only during boring meetings but also, and strikingly, while performing extremely simple tasks, including knocking

on a door. He also has the habit of disrupting meetings with his loud snoring.

The Pickwickian syndrome, a combination of obstructive apnea in moderately to severely obese people, most of whom attempt to sleep face up, takes its name from the famous Dickens description of Joe in his *Pickwick Papers*, so indicated in the classic paper by Burwell and co-workers [2]. The XIX century “case report” of the “wonderfully fat boy” Joe has become so diffused and acknowledged that many people now believe that Dickens was the first to furnish a description of this clinical picture.

On the contrary, just to limit the field to western medicine, various historical sources indicate the possibility of descriptions – even if not defined as such – of a syndrome related to sleep apnea well before the Fat Joe of Charles Dickens.

In the so-called Hippocratic Corpus (V–IV century BC) many hints and clues regarding apnea and sleepiness are present, and in a well recounted form too; the problem is that the interest of the author(s) of the different texts of the Corpus is focussed, on the one hand, on the relationship between apnea and sudden death, and, on the otherhand, on the relationship between sleepiness and epilepsy, and not specifically on sleep disturbances linked to respiratory alterations [3]. Therefore, a clear indication of sleep apnea as such is not exactly detectable, even if de facto present.

Some centuries later Pliny the Younger (61–112 AD), a Roman writer and scientist, while describing the eruption of Vesuvius (79 AD), also reported the death of his uncle, who “... rested, and gave every indication of actually sleeping; people who passed by his door heard his snores, which were rather resonant since he was a heavy man”. In the fine description by Pliny the Younger, obesity, sleepiness and snoring are presented in a unique clinical picture leading to the death of the single individual [4].

Today we know that obesity is a risk factor for “obstructive sleep apnea syndrome” (OSAS), and sleepiness and snoring are characteristic elements of this clinical, and now well-defined, syndrome. In fact, in 1965 the so-called Pickwickian syndromes, or obesity hypoventilation syndromes, were nosologically separated from OSAS [5]. Obstructive sleep apnea syndrome will be the topic discussed in this paper.

Current definition, epidemiology, predisposing factors and prognosis of OSAS

The obstructive sleep apnea syndrome is a clinical condition characterized by the coexistence of

irregular breathing at night with excessive daytime sleepiness, and it represents a major social health issue because of its high prevalence and of growing public awareness of it [6]. OSAS must always be distinguished from the central sleep apnea syndrome, whose mechanisms are strictly linked to alterations in the central nervous system, and which is therefore a condition pathogenically different from OSAS, even if the two separate pathogenic pictures largely overlap on clinical grounds.

When the sufferer of OSAS falls asleep, the muscle tone in the upper pharyngeal airway decreases. This determines a narrowing of the upper airway, leading to an increase in inspiratory effort in turn provoking a transient arousal or a lighter sleep phase that allows restoration of the normal calibre and tone of the airways. After this phase, the patient falls asleep again and the full cycle repeats itself even many hundreds of times during the same night, thus determining hypoxaemia, a fragmentation of normal sleep architecture and a depression of the quality of sleep. Consequently, features of OSAS include snoring, impaired concentration, restless sleep, choking episodes during sleep, irritability, nocturia [7]. The frequency of apneas and hypopneas is adopted to evaluate the severity of the OSAS and is called the apnea/hypopnea index (AHI), and OSAS may therefore be subdivided into varying degrees of breathing disorder depending on the AHI. Although any cut off in AHI, aimed at attempting to stratify the severity of OSAS, is arbitrary, a currently diffused classification distinguishes the severity of the condition in mild (if the AHI is 5–14 per hour), moderate (if the AHI is between 15 and 30 per hour), and severe (if it is higher than 30 per hour).

The Wisconsin Sleep Cohort Study has documented a prevalence of five episodes of apnea/hypopnea per hour in 24% of male individuals and in 9% of women. In this same study, conducted on approximately 1000 subjects, 4% of men and 2% of women had excessive sleepiness and a AHI higher than 5 [8]. Since differences in prevalence, depending on the ethnic origin of various populations are present, a conservative overall estimate needs to be drawn. Therefore, in summary, in middle-aged men the prevalence of OSAS can be estimated in the range 0.5–3%, and the majority of studies provides a prevalence of 1–2%, which is, to put this public health issue into perspective, twice the prevalence of severe asthma. A conservative estimate of the prevalence of OSAS in middle-aged women is in the range of 0.5–1% [7].

An increased risk of OSAS is associated with disorders predisposing to the stenosis of the upper airways and to the reduction of their stability [6].

Among these may be cited in particular the presence of obesity, a neck circumference higher than 43 cm, an abnormally small mandible size, reduction of nasal patency, obstruction of the upper airway due to any cause, macroglossia, and increased tonsillar size. Other elements predisposing to apneas are male gender, increasing age, sedative drugs, smoking and alcohol use. Recently a relationship between the sleep apnea syndrome and depression has been suggested in the scientific literature [9].

Even if this last putative association deserves further investigation, and the real incidence of depression in OSAS patients needs to be clarified, the long term prognosis for individuals affected by severe-degree not-treated OSAS is in any case negative in terms of a worsening quality of life, deteriorating cognitive functions, a consistent probability of car accidents, a higher incidence of hypertension and cardiovascular diseases and general, premature mortality [6].

Recent evidence on the relationship between OSAS and cardiovascular diseases

The association between cardiovascular diseases and OSAS has been suggested in recent times; the hypothesized pathway between cardiovascular disorders and sleep-disordered breathing is complex and seems to be bi-directional. In effect, the documented association may be mediated by higher cardiovascular risk factor levels in individuals with OSAS [10].

With specific regard to arterial hypertension, the above-mentioned bi-directional pathway is evident. Observational studies sustain an independent strong association between OSAS and hypertension, which worsens in a parallel way to the worsening of OSAS, as confirmed by an odds ratio of 1.2 for a mild degree OSAS and an odds ratio of 3.1 for a serious form of OSAS. Patients with OSAS have significantly higher blood pressure values than matched controls, and epidemiological studies have documented that the presence of OSAS is an independent predictor of augmented blood pressure, even after adjustment for known confounding variables. Appropriate treatment of OSAS (continuous positive airway pressure) can reduce blood pressure by up to 3.3 mm Hg over 24 h, with the highest decrease in people with the highest nocturnal hypoxaemia, in whom the mean 24 h fall in diastolic blood pressure can be as high as 5 mm Hg. A reduction in blood pressure of this magnitude can

decrease cerebral stroke risk by approximately 40% over a five to ten year period [7]. The extent of the efficacy of this blood pressure lowering intervention provides indirect evidence of the relevant prognostic role of arterial hypertension in the obstructive sleep apnea syndrome.

Distinguished characters of the past potentially affected by OSAS: has OSAS been a clinical condition typical of United States Presidents?

To remain in the field of hypertension, this was first documented in President Franklin D. Roosevelt (1882–1945) when he was 48 years old. He also suffered of coronary artery disease and heart failure. Historical sources indicate that this President had several predisposing factors and multiple characteristic elements of OSAS: in effect, apart from hypertension and cardiovascular disease, he smoked, he snored and he had a history of poliomyelitis. During the last phase of his life he apparently seemed depressed and “strangely” tired, even in the morning hours. Many times he complained of his excessive daytime sleeping, and occasionally he nodded off during simple conversations [11]. A posteriori reconstruction of the clinical picture of a figure of the past is always difficult and complex. However, it is perfectly possible that, towards the end of his life, on the one hand, President Roosevelt presented central sleep apnea as a consequence of his heart failure; on the other, a number of elements suggest the presence of an obstructive peripheral sleep apnea syndrome, including multiple predisposing factors for OSAS, as well as specific features of the patient’s clinical picture and the pattern of his death.

Other United States Presidents have been alleged to be sufferers of OSAS. Recently JG Sotos has provided convincing evidence that President William Howard Taft (1857–1930) was affected by the obstructive sleep apnea syndrome [12]. The massive central obesity of Taft is well known: he had a body mass index of at least 42 (current standard for obesity definition foresees a body mass index of 30 kg/m²), had a short generous neck, had exaggerated daytime somnolence, presented snoring, was hypertensive and, at least in the last years of his life, manifested probable cognitive and psychosocial impairment. Moreover, Taft lacked the typical signs and symptoms of other syndromes causing excessive somnolence during day hours. In effect, he had none of the elements, such as pedal edema and cyanosis, indicating

possible diagnoses of narcolepsy, of periodic leg movements and of the obesity hypoventilation syndrome. Despite of this evidence, different biographers have wondered whether Taft had other pathological conditions; in our view, however, the fact that his somnolence was initially distressing, involuntary and correlated with his weight provides evidence in favour of an obstructive sleep apnea syndrome.

President Theodore Roosevelt (1858–1919) was always a great snorer. In the last years of his life T Roosevelt's weight increased significantly, and ultimately he died in a sudden, unexpected and premature (he was 61) way while sleeping [13]. Of course, pulmonary embolism and other diseases cannot be ruled out, but the association with snoring and obesity and the pattern of Roosevelt's death may imply an obstructive sleep apnea syndrome.

To remain in the XIX century, other distinguished characters are here proposed as patients potentially suffering of obstructive sleep apnea syndrome.

Grover Cleveland (1837–1908), another US President, both presented elements predisposing to the OSAS, including obesity, a short thick neck and alcohol consumption, and manifested typical signs of OSAS, particularly snoring. President Cleveland wrote few personal letters, or at least few letters are available to us today, and therefore additional elements are difficult to find for the definition of a more complete historical pathological picture [14]. However, in this obese XIX century male too, various suggesting factors for OSAS are present.

The portraits of Napoleon Bonaparte (1769–1821) available in the Hotel des Invalides in Paris clearly document the striking difference between Napoleon as a young man (in 1802, when he was 33 years old) and a mature, defeated man (in 1814). In the 1814 portrait, the emperor appears as a tired and obese man, with a short thick neck [15]. Literary sources indicate that he had nasal obstruction and that he slept a lot during the day in his last years of life, also showing a decline in intellectual functions. The various elements predisposing towards OSAS, here recalled [16], are convincing, even if confounding factors with regard to the sleep pattern of Napoleon must be recognized. In effect, it is well known that Bonaparte was in the habit of sleeping a very limited number of hours per night from the time he was a young man, often waking up in the middle of the night to work. He was able to sleep anywhere and in any time, and no documentation of his snoring is available in historical sources. The diagnostic hypothesis of OSAS for Napoleon should neverthe-

less be considered as an integrative explanation of his declining physical and mental conditions in late life, since the multiple hypotheses so far advanced to motivate his health status (epilepsy? Pulmonary tuberculosis?) and even his death (poisoning? Peritonitis from gastric perforation?) are not completely convincing [17].

As already stated, the prevalence of OSAS in middle-aged men (0.5–3%) is currently higher than that in women of the same age (0.5–1%), and no available evidence indicates that in the past this epidemiological picture was different [7]. Therefore it should not surprise that the distinguished figures of the past examined in this text – up to this point – should all be men. However, a very important female personage of the XIX century can be a posteriori assessed, and adjudicated, for obstructive sleep apnea syndrome. We refer to Victoria (1819–1901), Queen of England, who had a waist measurement of around 46 in. and who, in later life, was notably obese and often fell asleep during the day. Records report that she suffered from a strange type of insomnia, and this could explain the daily sleepiness, the constant irritability and the memory loss that she more and more manifested from middle age on [18]. However, these same elements are also commonly associated with (obstructive) sleep apnea (syndrome), which in addition may masquerade itself as insomnia, and the contemporary presence of obesity and of a thick neck in a lady 5 ft 2 in. high (the height was measured when she was a young woman) reinforces the probability of OSAS, even in a female subject.

Conclusions

The obstructive sleep apnea syndrome, defined as the coexistence of excessive daytime sleepiness associated with irregular breathing at night, accompanied by impaired concentration, should be suspected in all patients who are hypertensive, obese, hypersomnolent and habitual snorers [19].

In the light of its current definition and of its major features, a number of famous people of the past could be suspected to have suffered from OSAS, as discussed in the present contribution. The possible presence of this clinical condition in such personages may have influenced not only their health status, but also their social life and their working activity, considering the physical, psychological and cognitive consequences of the presence and seriousness of the syndrome itself [20]. In effect, nowadays it is well known that OSAS can have practical unfavourable consequences on the daily life of every individual affected; for example, the

presence of this syndrome increases per se the risk of car accident by 3–7 times [6].

If at this point no doubt concerning the potential and real seriousness of OSAS should remain, a doubt still remains with regard to the presence of OSAS in famous historical figures. Just to furnish a further, final enigma: could the car accidents of Sir Winston Churchill (1874–1965), a well known obese, an alcohol and cigars estimator, and great snorer, be attributed to the obstructive sleep apnea syndrome? Time may never tell.

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References

- [1] Dickens C. *The Pickwick Papers*. Penguin Classics; 1961.
- [2] Burwell CS, Robin ED, Whaley RD, et al. Extreme obesity associated with alveolar hypoventilation: a Pickwickian syndrome. *Am J Med* 1956;21:811–8.
- [3] Diller H. *Hippocratis De aere aquis locis*. Corpus Medicorum Graecorum. Berlin; 1970.
- [4] Pliny the Younger. Letter to Tacitus 6.16.
- [5] Jung R, Kuhlo W. Neurophysiological studies of abnormal night sleep and the Pickwickian syndrome. *Prog Brain Res* 1965;18:140–59.
- [6] Clinical evidence, vol. 15. BMJ Publishing Group; 2006.
- [7] Scottish Intercollegiate Guidelines Network (SIGN) 73. Management of obstructive sleep apnea/hypopnea syndrome in adults. A national clinical guideline. Edinburgh, June 2003.
- [8] Young T, Palta M, Dempsey J, Skatrud J, Weber S, Badr S. The occurrence of sleep-disordered breathing among middle-aged adults. *N Engl J Med* 1993;328:1230–5.
- [9] Andrews JG, Oei TPS. The roles of depression and anxiety in the understanding and treatment of obstructive sleep apnea syndrome. *Clin Psychol Rev* 2004;24:1031–49.
- [10] Newman AB, Nieto FJ, Guidry U, Lind BK, Redline S, Shahar E, et al. Relation of sleep-disordered breathing to cardiovascular disease risk factors. *Am J Epidemiol* 2001;154:50–9.
- [11] Ferrell RH. *The dying President: Franklin D. Roosevelt 1944–1945*. Columbia (MO): University of Missouri Press; 1998.
- [12] Sotos JG. Taft and Pickwick. Sleep apnea in the White House. *Chest* 2003;124:1133–42.
- [13] Davis OK. Released for Publication: Some Inside Political History of Theodore Roosevelt and his Times 1898–1918. Boston: Houghton Mifflin; 1925.
- [14] Brodsky A. *Grover Cleveland: a study in character*. NY: St. Martin's Press; 2000.
- [15] Polkey MI, Morrell MJ, Simonds AK. Apnea and history. *Chest* 2004;125:1587.
- [16] Chouard C, Meyer B, Chabolle F. Did Napoleon suffer from sleep apnea syndrome? *Ann Otolaryngol Chir Cervicofac* 1988;105:299–303.
- [17] Camici M. Death of Napoleon. *Minerva Med* 2003;94:191–5.
- [18] Brewer C. *The death of Kings: a medical history of the Kings and Queens of England*. London: Abson Books; 2000.
- [19] Kryger MH. Sleep apnea: from the needles of Dionysius to continuous positive airway pressure. *Arch Intern Med* 1983;143:2301–3.
- [20] Strohl KP, Redline S. Recognition of obstructive sleep apnea. *Am J Respir Crit Care Med* 1996;154:279–89.

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